

Pablo Leyva

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EDUCATION

New Jersey Institute of Technology

Newark, NJ

B.S. Applied Mathematics & Computer Science

Expected: May 2027

Coursework: Data Structures, Machine Learning, Neurosymbolic AI, Explainable AI, Deep Learning, Reinforcement Learning

EXPERIENCE

Research Engineer Intern, *Mechanistic Interpretability* 

Sep 2026 – Present | Newark, NJ

- Advised by Kieran Murphy to build a **PyTorch** pipeline classifying all **144 GPT-2 attention heads** by cross-checking in **TransformerLens** and **Hugging Face** to quantify no-op attention behavior; 74% default to the BOS sink.
- Re-ran the 3,000 sequences of rare, common and byte-free tokens; 15 heads shifted oppositely in early and late layers.
- Running **causal experiments** of 1.2M sequences to isolate attention shift.

AI/ML Intern, *Apple*

May 2026 – Aug 2026 | Cupertino, CA

- Designed agentic-commerce use cases for Siri, using Apple Wallet and Event Passes data as context to anticipate user needs.
- Built a **multimodal synthesizer** fusing speech and **On-Screen Awareness** into instruction prompts.
- Proposed a screenless human-computer interface that delegates work to recursive subagents on the user's behalf.

Machine Learning Intern, *Angeles Investors*

Jan 2026 – May 2026 | New York City, NY

- Productized AI **agent swarm** powered by GPT-4o which led to a **\$500K pre-seed deal** at a \$48M pre-money valuation.
- Deployed an **asynchronous FastAPI** data ingestion pipeline via **Docker** that streamed structured results in real time, accelerating startup screening, due diligence and outreach.

President, *Society of Hispanic Professional Engineers (SHPE)*

Sep 2025 – May 2026 | Newark, NJ

- Led 13 officers fundraising **\$40,000** across **90+** events year-round, growing membership to a record **250+**.
- Architected and shipped the chapter's **first iOS app** in **27 days** leading 3 engineers, scaling to **200+ users** in its **first month** with event registration, member details, and mentorship-program features.

AI/ML Intern, *Apple*

May 2025 – Aug 2025 | Cupertino, CA

- Deployed a **LangGraph** agent integrating **RAG** with **MCP** to reason over technical docs for Apple Pay Merchants.
- Used **Qdrant** for **multilingual similarity search** and **GraphQL** to fetch and manipulate data across internal systems.
- Built **LLM evaluation pipeline** benchmarking agent trajectories, tool-use success, groundedness, and hallucination rate.

PROJECTS

Evolv | [Devpost](#)

Flask, NumPy, Pandas, Python

- Placed **1st of 546** teams at HackPrinceton, building an entire backend and predictive modeling system in under 12 hours.
- Built a **geospatial stochastic simulation** platform to forecast longterm regional impact.
- Productionized a **real-time Kubernetes pipeline** with LLM-generated reports, end-to-end running under 5 seconds.

Founder - HAI

Transformers, PyTorch, pgvector, Redis, Docker, FastAPI, LangGraph, Python

- Commercialized an AI-powered SAT prep platform, selling to NJ tutoring centers supporting **600 students** concurrently.
- Built **multimodal** input/output **pipelines** for PDFs, images, and text, combining **OCR**, vector search, semantic retrieval, prompt orchestration, and GPT-4o API to produce personalized questions, explanations, and study material.
- Implemented a **symbolic verifier** for mathematical reasoning on CoT yielding **100%** solution correctness before delivery.

NeuroCache: KV-Cache Memory Manager | [GitHub](#)

PyTorch, ONNX, llama.cpp, Transformers.js, C++

- Implemented a KV-cache branching for **llama.cpp** with attention-scored eviction and slot-based allocation.
- Designed **cache-replacement algorithms** balancing memory footprint against response quality for long-context sessions.
- Benchmarked eviction strategies across context lengths, quantifying memory-efficiency and generation-quality tradeoffs.

Scholastic-LLM: On-Device Distillation | [GitHub](#) | [Hugging Face](#)

MLX, PEFT, Hugging Face, Python

- Distilled **Claude Sonnet 4.6** into an **8-bit Qwen 2.5 7B** via on-device **LoRA** in 12.7 GB, lifting task score **258%**.
- Built evals over 5 checkpoints; traced **DPO's** zero gain to preference saturation; shipped best **SFT** via **PEFT**.

RedLoop: Automated Red-Teaming | [GitHub](#)

asyncio, Ollama, Gemma 4, Python

- Built an **automated red-teaming** harness (**132** attempts across **jailbreak** and **agentic prompt-injection**) that found a **reproducible** injection hijacking a **tool-using agent** into an unauthorized action.
- Implemented a **PAIR-style adaptive attacker** over **LLM-judge** feedback
- Reported **attack-success-rate** with **95% CIs** by category, technique, and scenario to expose evaluation-coverage gaps

SKILLS

Languages: Python, C++, JavaScript/TypeScript, Swift

AI/ML: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LangGraph, RAG, MCP, Qdrant, llama.cpp